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UNIVERSITY OF BIRMINGHAM

School of Mathematics

Programmes in the School of Mathematics	Second Examination
Programmes in the School of Mathematics	Final Examination
Programmes involving Mathematics	Second Examination
Programmes involving Mathematics	Final Examination

2MVA 06 25667 Level I

LI Multivariable & Vector Analysis

2MVA3 06 35172 Level H

LH Multivariable & Vector Analysis

Alternative Assessment

January Examinations 2021-22

Three Hours

Full marks will be obtained with complete answers to all FOUR questions. Each question carries equal weight. You are advised to initially spend no more than 45 minutes on each question and then to return to any incomplete questions if you have time at the end. An indication of the number of marks allocated to parts of questions is shown in square brackets.

Section A

1. (a) Suppose that the positive real-valued function $z = z(x,y)$ is defined implicitly by the following equation

$$\cos(xy) + \ln(z) = yz - 1/2.$$

Find the partial derivatives z_x and z_y in terms of x, y and z. Further, calculate their values at the point $(x,y,z) = (\pi, 1/2, 1)$. [7]

(b) Write the equation for the tangent line to the intersection of the following two surfaces:

$$xy + yz + zx = 11 \text{ and } xyz = x + y + z$$

at the point $\mathbf{a} = (1,2,3)$. [9]

(c) Using the method of Lagrange multipliers, find the maximum and minimum values of the function

$$f(x,y,z) = x + z,$$

where (x,y,z) lies on the surface determined by the equation

$$x^2 + y^2 + z^2 = 1.$$

[9]

2. (a) (i) Give the definition of the cross product of two vectors, $\vec{a} = (a_1, a_2, a_3)$ and $\vec{b} = (b_1, b_2, b_3)$.

(ii) Prove that $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$.

[5]

(b) Prove that the integral $I = \int_C y^3 \cos 3x\, dx + y^2 \sin 3x\, dy$ is path-independent. [6]

(c) Find the work of the vector field $\vec{F} : \mathbb{R}^2 \to \mathbb{R}^2$, where $\vec{F}(x,y) = \left(-\frac{\pi}{y}, \frac{2\pi}{x}\right)$, along the arc, given by the section of the unit circle that starts at the point $(0,1)$ and ends at the point $(-1,0)$. [8]

(d) (i) Calculate the divergence of the following vector field:

$$\vec{F} \; = \; \left(4xz, \; -y^2, \; yz\right).$$

(ii) Find the curl of the following vector field:

$$\vec{F} \; = \; \left(x^2 + yz, \; y^2 + zx, \; z^2 + xy\right)$$

at the point $(1,2,3)$.

[6]

Section B

3. Consider the triple integral

$$I = \iiint\limits_S z\,dxdydz,$$

where the domain S lies in the first octant of the real space, i.e. where x, y and z are non-negative, and is bounded below by the surface $z = \left(x^2 + y^2\right)^2$ and bounded above by the surface $x^2 + y^2 + z^2 = 2$.

(a) Calculate the integral I using cylindrical coordinates. [10]

(b) Express the integral I as repeated integrals using Cartesian coordinates. [3]

(c) Express the integral I as repeated integrals using spherical coordinates. [6]

(d) Using the result of (3a), or otherwise, calculate the following integral

$$J = \iiint\limits_T \left(z + x\sin(y^2)\right) dxdydz,$$

where the domain T is enclosed by the two surfaces $z = \left(x^2 + y^2\right)^2$ and $x^2 + y^2 + z^2 = 2$.

[6]

4. (a) Find the value of the integral $I = \int_C \left(2x + \frac{1}{y}\right) dx + \left(2y - \frac{x}{y^2}\right) dy$ along an arbitrary smooth curve C, which starts at the point $A(1,1)$ and ends at the point $B(2,2)$ without crossing the x-axis. [15]

(b) Compute the value of the line integral

$$\oint_C \left(x^2 + y\right) dx + 2x\,dy$$

using Green's theorem, where C is the counterclockwise-oriented triangle with vertices $(0,0)$, $(1,0)$ and $(0,1)$. In your solution please include an explanation as to why Green's theorem can be applied here. [10]